

DARSH SRIVASTAVA

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GitHub: github.com/darshsrivastava234-byte/Embedded-IoT-simulation-

Professional Summary

Electronics and Communication Engineering undergraduate with a strong foundation in **Embedded Systems, Digital Electronics, Analog Electronics and IoT**. Experienced in developing projects using **ESP32, Arduino, Verilog HDL, and Tinkercad**. Skilled in **Embedded C/C++, debugging, and hardware-software integration**, with a passion for building practical engineering solutions.

Technical Skills

- Languages: Embedded C, C++, Java, Verilog HDL
- Embedded Systems: ESP32, Arduino, Microcontrollers, Digital Electronics, IoT
- Protocols: Wi-Fi, BLE, MQTT
- Tools: Arduino IDE, Tinkercad, Git, VS Code, ROS
- Soft Skills: Problem Solving, Analytical Thinking, Teamwork, Communication

Education

- **Bachelor of Technology (B.Tech) – Electronics and Communication Engineering VIT Bhopal University** | Current CGPA: 7.81/10
- **Class 12 (PCM)** – Dr. Virendra Swarup Education Centre, Avadhपुरi, Kanpur Percentage: 83.6%
- **Class 10** – Dr. Virendra Swarup Education Centre, Avadhपुरi, Kanpur Percentage: 93%

Internship

Maven Silicon – VLSI Design Intern

- Worked on **APB-based GPIOs project**, implementing register transfer logic and verifying functionality.
- Gained exposure to **VLSI design flow, bus protocols, and digital verification techniques**.

Projects

1). 15-Day Embedded Systems & IoT Series (Arduino/ESP32)

- Simulated and developed 15 projects including:
- Bionic Arm Simulation
- Automatic Plant Watering System
- Electronic Voting Machine (EVM)
- Password-Protected Door Lock System
- Home Automation System using Arduino Uno
- Digital Clock
- Dual Axis Solar Tracker
- Motion-Controlled Colour Changer
- Smart Greenhouse Automation System
- Tyrannosaurus Rex Dino Game
- Obstruction Avoidance Robot
- 4×4×4 LED Cube
- Calculator Using Arduino
- Pac-Man Animation
- Visual Pattern Animation

2). Robotic Arm (Gesture + Dial Control)

Built a dual-mode robotic arm integrating **gesture recognition** and **dial-based task execution**. Demonstrated real-time synchronization between sensors and actuators.

3). Remote-Controlled Water Pollution Monitoring Boat

Designed a multi-stage embedded system for **environmental monitoring and waste collection**. Proposed sensor-based control for efficient water body management.